$$Na + C + N + S \xrightarrow{Heat} \underset{\text{sodium sulphocyanide}}{NaCNS}$$

$$3NaCNS + FeCl_3 \longrightarrow \underset{\substack{\text{ferric sulphocyanide}\\ \text{(Blood red colour)}}}{Fe(CNS)_3} + 3NaCl$$

iii) Test for sulphur:

a) To a portion of the lassaigne's extract, add freshly prepared sodium nitro prusside solution. A deep violet or purple colouration is obtained. This test is also used to detect S^{2-} in inorganic salt analysis

$$Na_2S + Na_2[Fe(CN_5)NO] \rightarrow Na4[Fe(CN_5)NOS]$$

sodium nitro prusside

b) Acidify another portion of lassaigne's extract with acetic acid and add lead acetate solution. A black precipitate is obtained.

$$(CH_3COO)_2Pb + Na_2S \longrightarrow \underset{\text{(black ppt)}}{PbS\downarrow} + 2CH_3COONa$$

c) **Oxidation test:** The organic substances are fused with a mixture of KNO_3 and Na_2CO_3. The sulphur, if present is oxidized to sulphate.

$$Na_2CO_3 + S + 3O \longrightarrow Na_2SO_4 + CO_2$$

The fused mass is extracted with water, acidified with HCl and then $BaCl_2$ solution is added to it. A white precipitate indicates the presence of sulphur.

$$BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$$

iv) Test for halogens: To another portion of the lassaigne's filtrate add dil HNO_3 warm gently and add $AgNO_3$ solution.

a) Appearance of curdy white precipitate soluble in ammonia solution indicates the presence of chlorine.

b) Appearance of pale yellow precipitate sparingly soluble in ammonia solution indicates the presence of bromine.

c) Appearance of a yellow precipitate insoluble in ammonia solution indicates the presence of iodine.

$$\underset{\substack{\text{from}\\ \text{organic}\\ \text{compound}}}{Na + X} \xrightarrow{heat} NaX \quad \text{(Where x= Cl, Br, I)}$$

$$Nax + AgNO_3 \longrightarrow AgX + NaNO_3$$

If N or S is present in the compound along with the halogen, we might obtain NaCN and Na_2S in the solution, which interfere with the detection of the halogen in the $AgNO_3$ test Therefore we boil the lassaignes extract with HNO_3 which decomposes NaCN and Na_2S as

$$NaCN + HNO_3 \xrightarrow{\Delta} NaNO_3 + HCN\uparrow$$

$$Na_2S + 2HNO_3 \xrightarrow{\Delta} 2NaNO_3 + H_2S\uparrow$$

further

$$NaCN + AgNO_3 \longrightarrow \underset{\text{white ppt confusing with AgCl}}{AgCN} + NaNO_3$$

$$Na_2S + AgNO_3 \longrightarrow \underset{\text{black ppt}}{Ag_2S\downarrow} + NaNO_3$$

V) Test for phosphorous: A solid compound is strongly heated with a mixture of Na_2CO_3 & KNO_3. phosphorous present in the compound is oxidized to sodium phosphate. The residue is extracted with water and boiled with Conc. HNO_3. A solution of ammonium molybdate is added to the above solution. A canary yellow coloration or precipitate shows the presence of phosphorous.

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